

Ishan Paharia

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EDUCATION

The LNM Institute of Information Technology

B.Tech in Computer Science, GPA: 8.35/10

Jaipur, India

2023 – 2027

PROJECTS

Pixnette — Real-Time Collaborative Pixel Canvas — Live Demo

React, Tailwind CSS, Node.js, Express, PostgreSQL (Neon), Socket.io, Redis, Docker, Nginx

- Engineered horizontal scaling across 2 Node.js instances behind an Nginx round-robin reverse proxy, sustaining 300 concurrent WebSocket connections on a self-hosted VPS.
- Built a Redis write-back pipeline with an ordered event queue, atomic RENAME flush locking, and timestamp-guarded UPSERTs to guarantee canvas consistency and full timeline history across bulk PostgreSQL flushes.
- Synchronized multi-instance Socket.io state via `@socket.io/redis-adapter` Pub/Sub and a binary Redis canvas buffer using `SETRANGE/GET` for $O(1)$ per-pixel read/write.
- Hardened deployment with Redis TTL cooldowns, UUID/IP-UA fingerprinting, Nginx rate limits on WebSocket and API routes, and GitHub Actions CI/CD.

CoursessionAI — AI Learning Platform — Live Demo

React, Tailwind CSS, Node.js, Express, PostgreSQL (Neon)

- Built a full-stack platform that converts any YouTube playlist into a structured course with AI-generated module hierarchies, per-video summaries, quizzes, and a context-aware AI tutor.
- Engineered an atomic playlist ingestion pipeline using Neon's WebSocket client for multi-step transactional DB writes, eliminating partial-state corruption across course, module, and video inserts.
- Implemented layered access control — Clerk JWT verification, user-level course ownership, and resource-level integrity checks — securing all progress, notes, bookmark, and AI endpoints.
- Deployed on Vercel with rate-limited APIs, Slix webhook verification via raw body capture, and XSS-sanitized completion certificate generation.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript

Frameworks: React, Next.js, Node.js, Express.js, Django, FastAPI, Tailwind CSS

Databases: MongoDB, PostgreSQL, MySQL, Redis

Tools & Technologies: Docker, WebSockets, Git, Linux/Bash

Others: Data Structures & Algorithms, OOP, DBMS

CO-CURRICULAR

Coordinator and Mentor

Aug 2024 – Apr 2026

Art and Craft Club (Vignette)

- Coordinated workshops and events for 100+ participants, handling end-to-end logistics and scheduling.
- Mentored junior members across semester-long creative projects, driving consistent participation.

ACHIEVEMENTS

Codeforces: Specialist (Max Rating: **1465**)

CodeChef: 3 Star Coder (Max Rating: **1746**)

LeetCode: Solved **200+** problems, (Max Rating: **1853**)